

SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26104
- **Problem Statement Title** – AI-Powered Real Time Detection and Prevention of Voice Cloning Impersonation Attacks
- **Theme** – Blockchain & Cybersecurity
- **PS Category** – Software
- **Team ID** – 041
- **Team Name** – The Grenades





VOICEGUARD

Real-Time Voice Cloning Detection and Authentication System

PROBLEM

- **AI Can Mimic Human Voices:** Cloned voices can closely imitate real speakers.
- **Impersonation Risk:** Fake voices can be used to impersonate trusted individuals.
- **Late Detection:** Existing methods may detect fake audio only after recording.

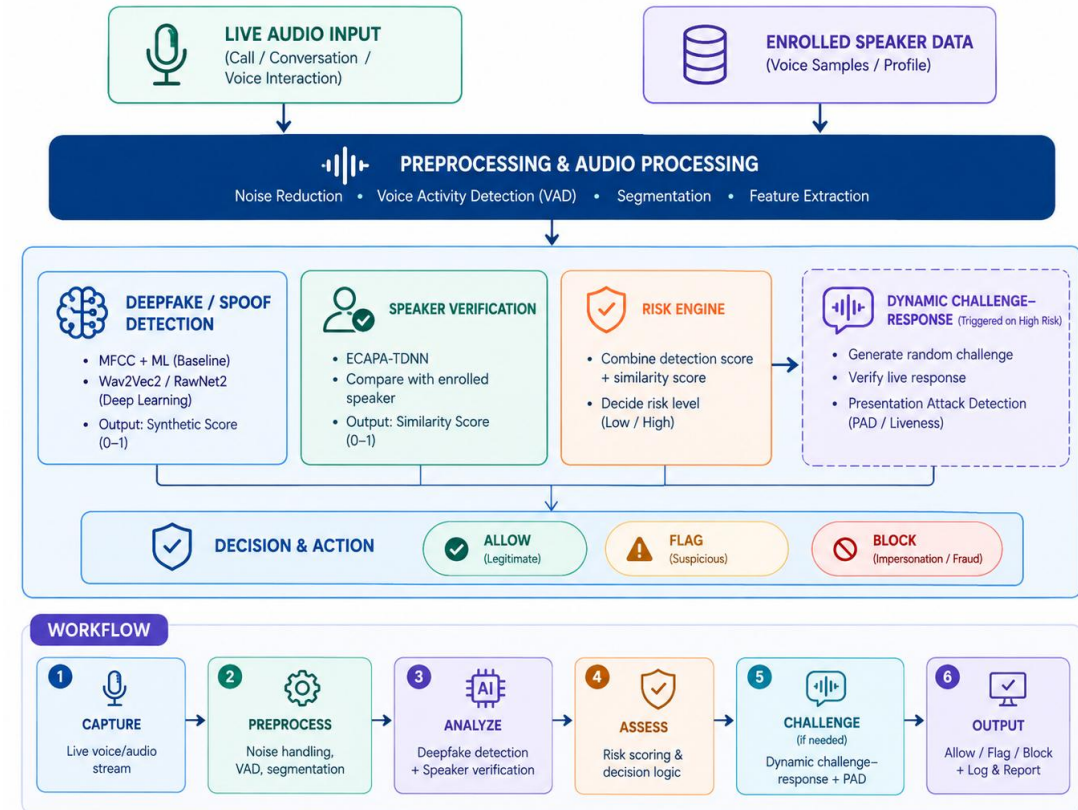
PROPOSED SOLUTION

- **Real-Time Detection:** Continuously analyze speech for AI-generated or manipulated voices.
- **Multi-Layer Verification:** Combine voice deepfake detection with speaker verification.
- **Active Prevention:** Use liveness checks and dynamic challenges for high-risk interactions.

UVP (Unique Value Product)

- **Beyond Detection:** Detects suspicious voices and enables preventive action.
- **Multi-Layer Security:** Combines deepfake detection, speaker verification and liveness.
- **Real-Time & Adaptive:** Performs additional verification when risk is detected.

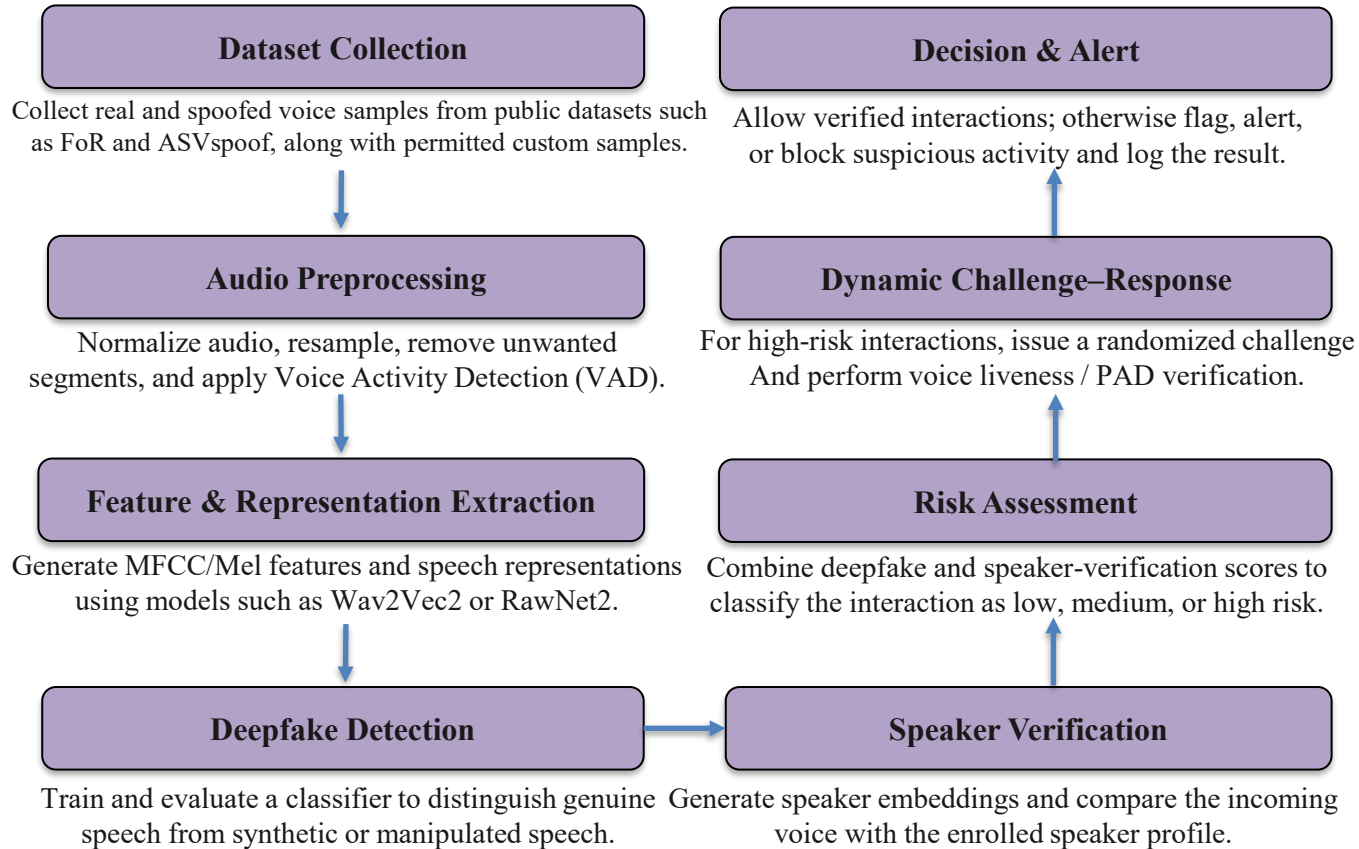
• SYSTEM ARCHITECTURE & WORKFLOW •





TECHNICAL APPROACH

METHODOLOGY



DATA → MODEL → VERIFICATION RISK → CHALLENGE → DECISION

WORKING PROTOTYPE FLOW



TECH STACK





FEASIBILITY AND VIABILITY

From Technical Possibility to Practical Deployment

FEASIBILITY



- Established AI models can support reliable voice analysis and verification.
- Public datasets provide sufficient data for initial model training and testing.
- Real-time processing is achievable using web audio capture and streaming.
- Modular components allow detection and verification to be developed independently.

POTENTIAL CHALLENGES AND RISKS



- Voice-cloning techniques evolve rapidly, making unseen attacks harder to detect.
- Background noise and poor audio quality can reduce detection accuracy.
- Real-time analysis requires low latency while maintaining model performance.
- False positives or negatives may lead to incorrect security decisions.
- Voice data requires careful handling to protect user privacy.

STRATEGIES FOR OVERCOMING THESE CHALLENGES



- **DIVERSE DATA** → Generalization
- **ROBUST PREPROCESS** → Noisy Audio
- **MODEL OPTIMIZATION** → Low Latency
- **MULTI-LAYER VERIFY** → Fewer False Decisions
- **SECURE DATA** → Privacy



IMPACT AND BENEFITS

IMPACT

Safer Voice Communication



Reduces risks of voice impersonation.

Stronger Identity Verification



Verifies speaker authenticity before access.

Real-Time Threat Detection



Detects suspicious voices during interactions.

Faster Fraud Response



Enables immediate alerts and verification.

Trust in Digital Voice Services



Improves confidence in voice-based services.

BENEFITS

Social Security



Protects users from voice-based impersonation.

Economic Protection



Helps reduce losses from voice fraud.

Scalable Security



Adapts across different voice applications.

Privacy-Aware Protection



Minimizes exposure of sensitive voice data.

Future-Ready Defense



Prepares against evolving voice-cloning threats.



RESEARCH AND REFERENCES



- ◆ **ASVspoof 5 — Speech Deepfake Detection & Spoofing-Robust Speaker Verification:**

Summary: A major benchmark focused on speech deepfake detection and spoofing-robust automatic speaker verification provides relevant datasets, attack scenarios and evaluation methodology for developing anti-spoofing systems. [\[SOURCE\]](#)

- ◆ **End-to-End Anti-Spoofing with RawNet2:**

Summary: A major benchmark focused on speech deepfake detection and spoofing-robust automatic speaker verification provides relevant datasets, attack scenarios and evaluation methodology for developing anti-spoofing systems. [\[SOURCE\]](#)

- ◆ **Fake-or-Real (FoR) — Speech Deepfake Dataset:**

Summary: A publicly available dataset containing real and synthetic speech samples, used for developing and evaluating audio deepfake detection models. [\[SOURCE\]](#)

- ◆ **ECAPA-TDNN for Speaker Verification:**

Summary: Introduces ECAPA-TDNN, a neural architecture for extracting speaker representations and performing speaker verification. It is relevant to the proposed identity-verification layer of the system. [\[SOURCE\]](#)

- ◆ **wav2vec 2.0 — Self-Supervised Speech Representation Learning:**

Summary: Introduces a self-supervised approach for learning powerful representations directly from speech audio. Such speech representations can support downstream tasks involving audio analysis and classification. [\[SOURCE\]](#)